MILITARY STANDARD

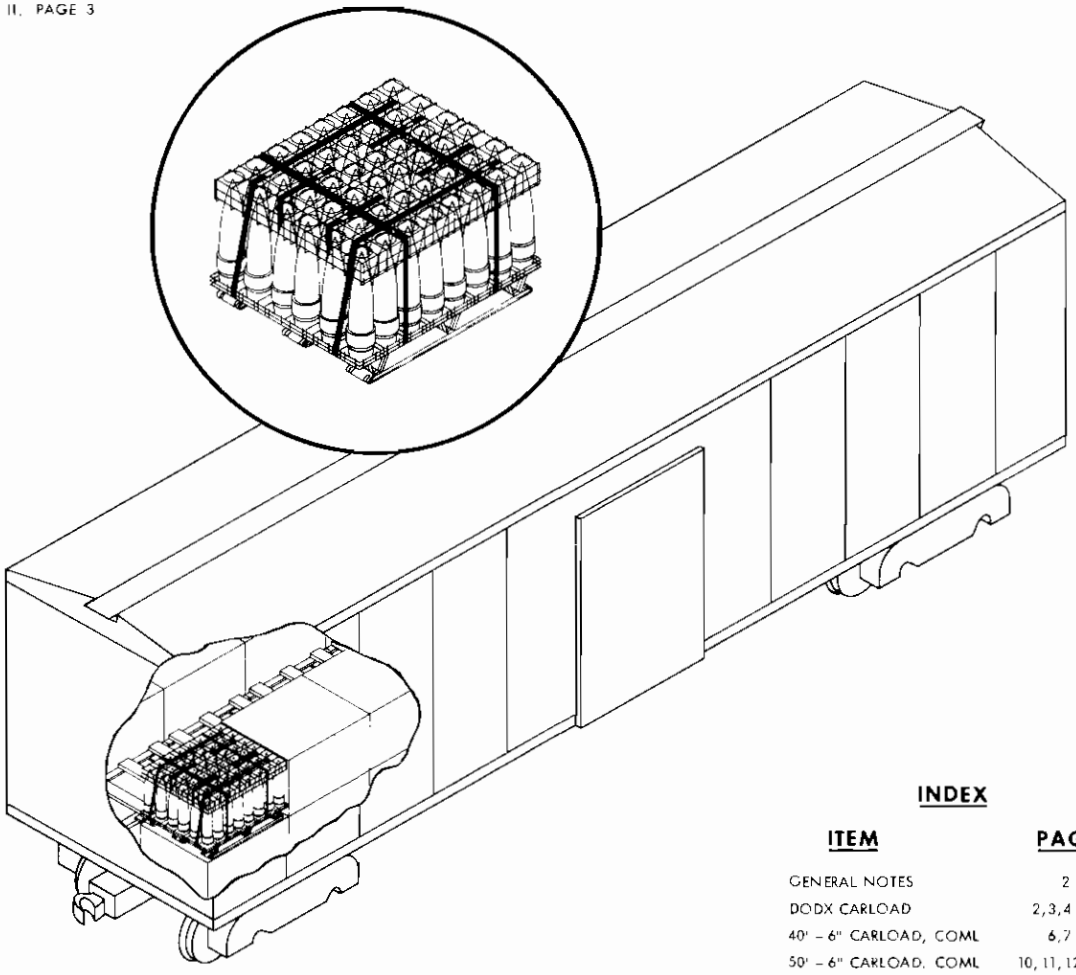
MIL-STD-1325-13
(NAVY)

27 MARCH 1973
SUPERSEDING
WR-52/13
23 MARCH 1965

CARLOADING
PROJECTILE, 5"/38 MKS 35 OR 49 & MODS
IN ADAPTER MK 11 & MODS
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING . . WR-54/4
SEE TABLE I, PAGE 2
DOT HAZARD CLASSIFICATION
SEE TABLE II, PAGE 3



INDEX

ITEM	PAGE
GENERAL NOTES	2
DODX CARLOAD	2,3,4 & 5
40' – 6" CARLOAD, COML	6,7 & 8
50' – 6" CARLOAD, COML	10, 11, 12 & 13

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

AUTHORIZED AND RELEASED FOR GENERAL USE.

M.S. Gray NWHL 3/26/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 3/26/73
SIGNATURE (ORD) SYSCOM BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

M. F. Grossmuck
DATE 3/6/73

ORIGINATOR L.R. Casinga
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF 5"/38 PROJECTILES MKS 35 OR 49 & MODS PALLETIZED IN ACCORDANCE WITH WR-54 4.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. APPLICABLE MATERIAL SPECIFICATIONS:

   DUNNAGE LUMBER - FED. SPEC. MM-L-751

   NAILS - FED. SPEC. FF-N-105

   STRAPPING - FED. SPEC. QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED)

   SEALS - FED. SPEC. QQ-S-781, STYLE III, HEAVY DUTY.
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL 1).

50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 32 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH (SEE MIL-STD-1325 (NAVY). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 3100 LBS OF THE LADING.

TABLE I

ITEM	UNIT LOAD DIMENSIONS			UNIT LOAD CUBE (CU. FT.)	UNIT LOAD WEIGHT (LBS) PALLET ADAPTER MK 11		CARLOAD WEIGHT (LBS) (INCLUDES 301 LBS DUNNAGE WT) PALLET ADAPTER MK 11	
	L	W	H		MOD 0	MOD 1	MOD 0	MOD 1
PROJECTILE WITHOUT WATERPROOF PROTECTING CAP	48	40	27 1/2	30.5	2965	2910	95,181	93,421
PROJECTILE WITH WATERPROOF PROTECTING CAP	48	40	28 3/8	31.6	3085	3030	99,021	97,261
PROJECTILE WITH WINDSHIELD	48	40	27 1/2	30.5	2965	2910	95,181	93,421

PIECE NO	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
9	BACK-UP STRIP	2 x 4 x 96	2	SEE 8	-	-
8	BUFFER STRIP	2 x 6 x 96	2	9	1 PER FOOT	10d
7	SEAL	3/4	12	-	-	-
6	STRAP	3/4 x .035 x 30	12	-	-	-
5	SPACER	2 x 4 1/8*** x 107	2	-	-	-
4	SWAY BRACE STOP	2 x 4 x 8	8	3	3	10d
3	CROSS BRACE	2 x 4 x W**	12	SEE 2	-	-
2	SWAY BRACE LONGITUDINAL	1 x 4 x 44	8	3	3 PER JOINT	10d*
1	SLEEPER	2 x 4 x 40	24	CAR FLOOR	2 PER FOOT	16d

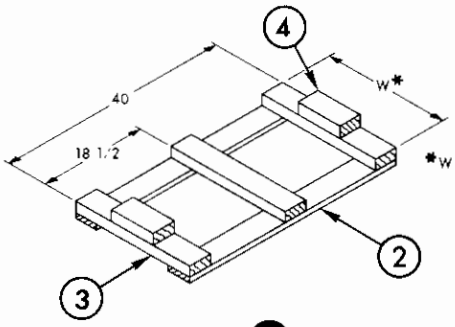
LIST OF MATERIALS & NAILING DATA

*CLINCHED

**W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR

***ACTUAL DIMENSION (CAN BE RIPPED FROM 2 x 6)

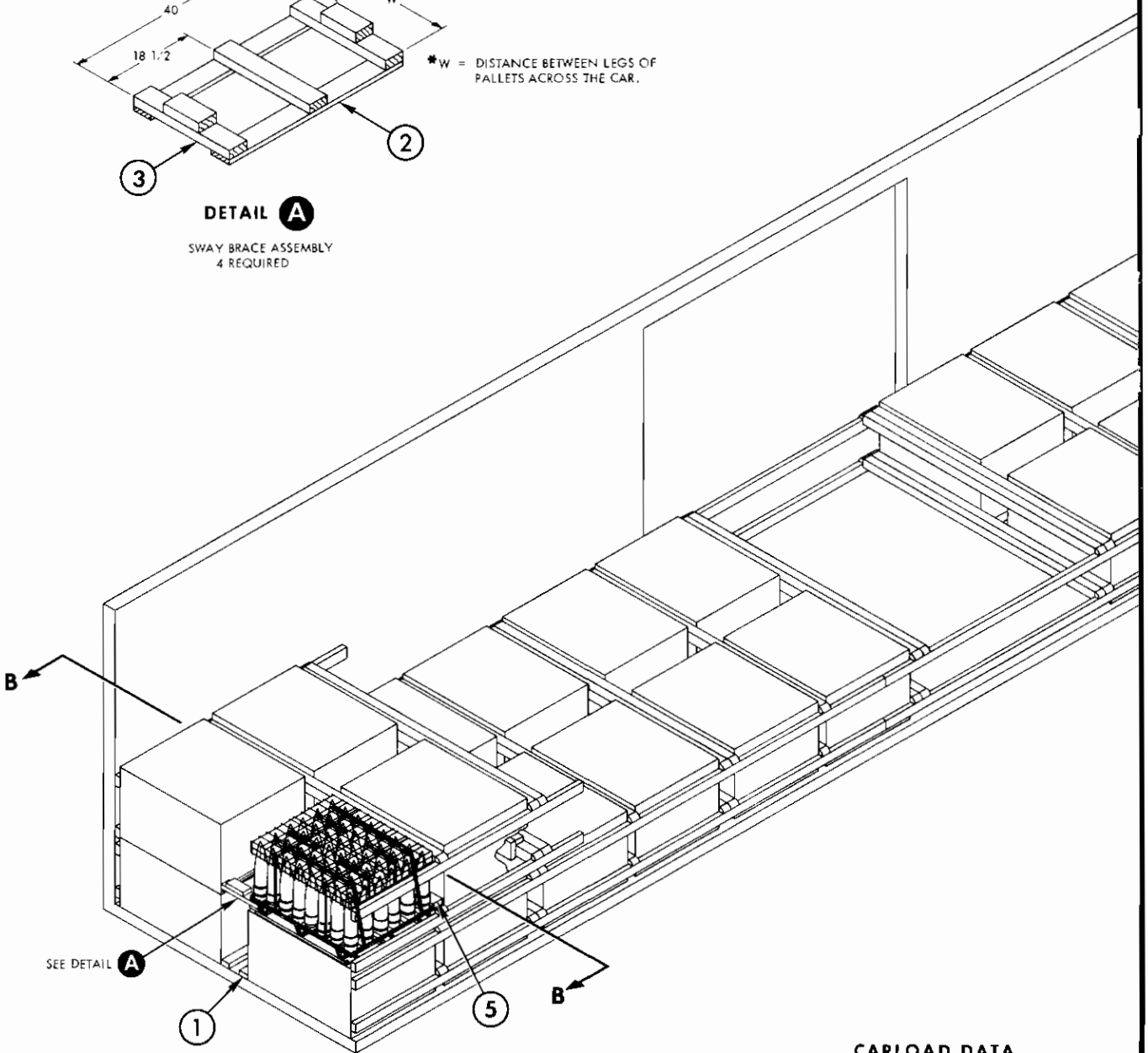
Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.



*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

DETAIL A

SWAY BRACE ASSEMBLY
4 REQUIRED



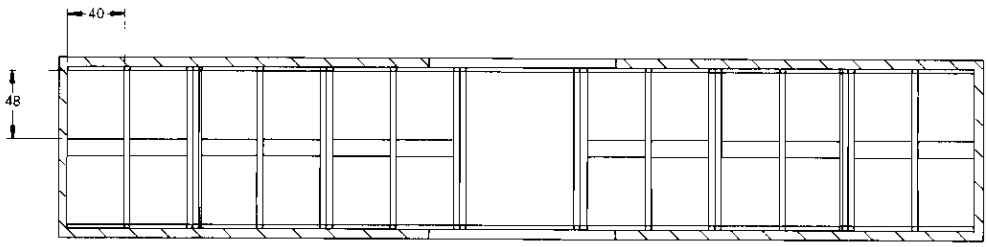
CARLOAD DATA

NUMBER OF UNIT LOADS 32
NUMBER OF CROSSMEMBERS REQ'D 48
CARLOAD WEIGHT . . . . . SEE TABLE 1

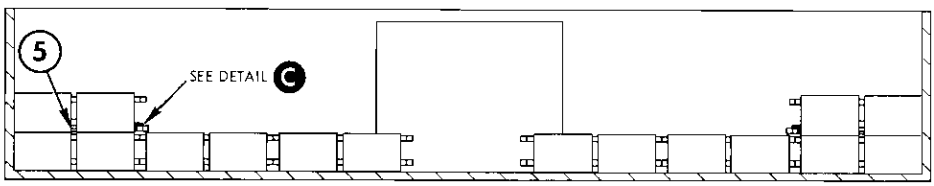
TABLE II

PROJECTILE	DOT HAZARD CLASSIFICATION
AAC	A
COM	A
FCLVT	A
FCLVTNF	B
FCLPDNF	B
HC	A
ILLUM	B
VTNF	B
VT	A
WP	A

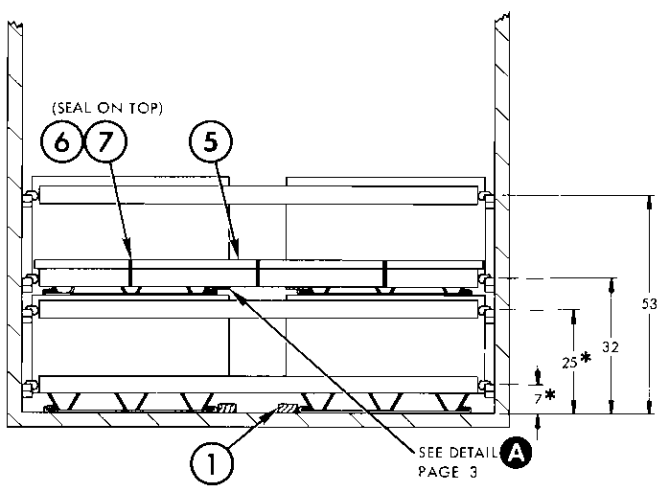
Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.



LOADING PLAN



ELEVATION

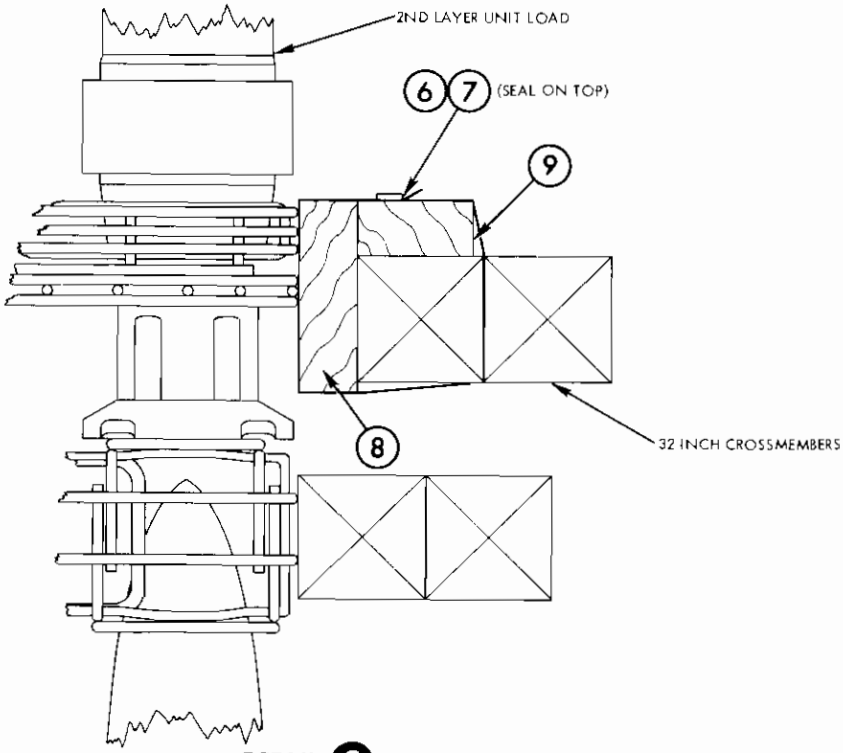


SECTION B-B

SHOWING LOCATION OF WALL MEMBERS
USED FOR CROSSMEMBERS

*DOORWAY MEMBER LOCATIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.



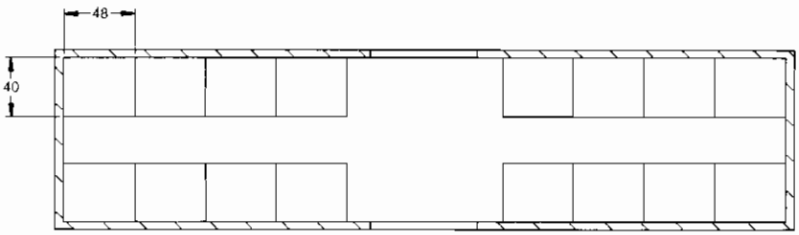
DETAIL C

SHOWING STRAPPING OF BUFFER STRIP & BACKUP STRIP TO CROSSMEMBER

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

40 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 32 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE, PROVIDED DOORWAY PROTECTION WHEN REQUIRED IS INSTALLED IN ACCORDANCE WITH WR-52.



LOADING PLAN

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
12	HOLDDOWN CLEAT	2 x 6 x 24	4	11	5	10d
11	GATE HOLDDOWN	2 x 6 x 10 FT	2	CAR WALL	5 EACH END	10d
10	VERTICAL TIE BAR	2 x 4 x 60	4	8	3 PER JOINT	16d
9	HORIZONTAL TIE BAR	2 x 4 x CAR WIDTH -1	4	8	3 PER JOINT	16d
8	STRUT	4 x 4 x WEDGE FIT***	16	6	2 PER JOINT	16d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH -1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 60	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH -1	8	6	3 PER JOINT	10d
4	CROSS BRACE	2 x 6 x W*	24	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 48	16	2, 4	SEE NOTE**	
2	SWAY BRACE SUPPORT	1 x 4 x (W + 8)*	16	SEE 3	-	-
1	SLEEPER	2 x 4 x 48	16	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS & NAILING DATA

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

**PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT.

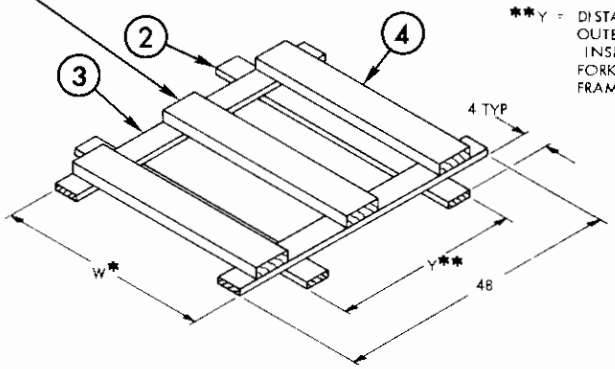
***2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4's.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

CROSS BRACES MUST BE CENTERED AGAINST LEGS OF PALLETS

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

**Y = DISTANCE BETWEEN INSIDE OF OUTER POSTS OF PALLET. INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.



DETAIL A

SWAY BRACE ASSEMBLY
8 REQUIRED

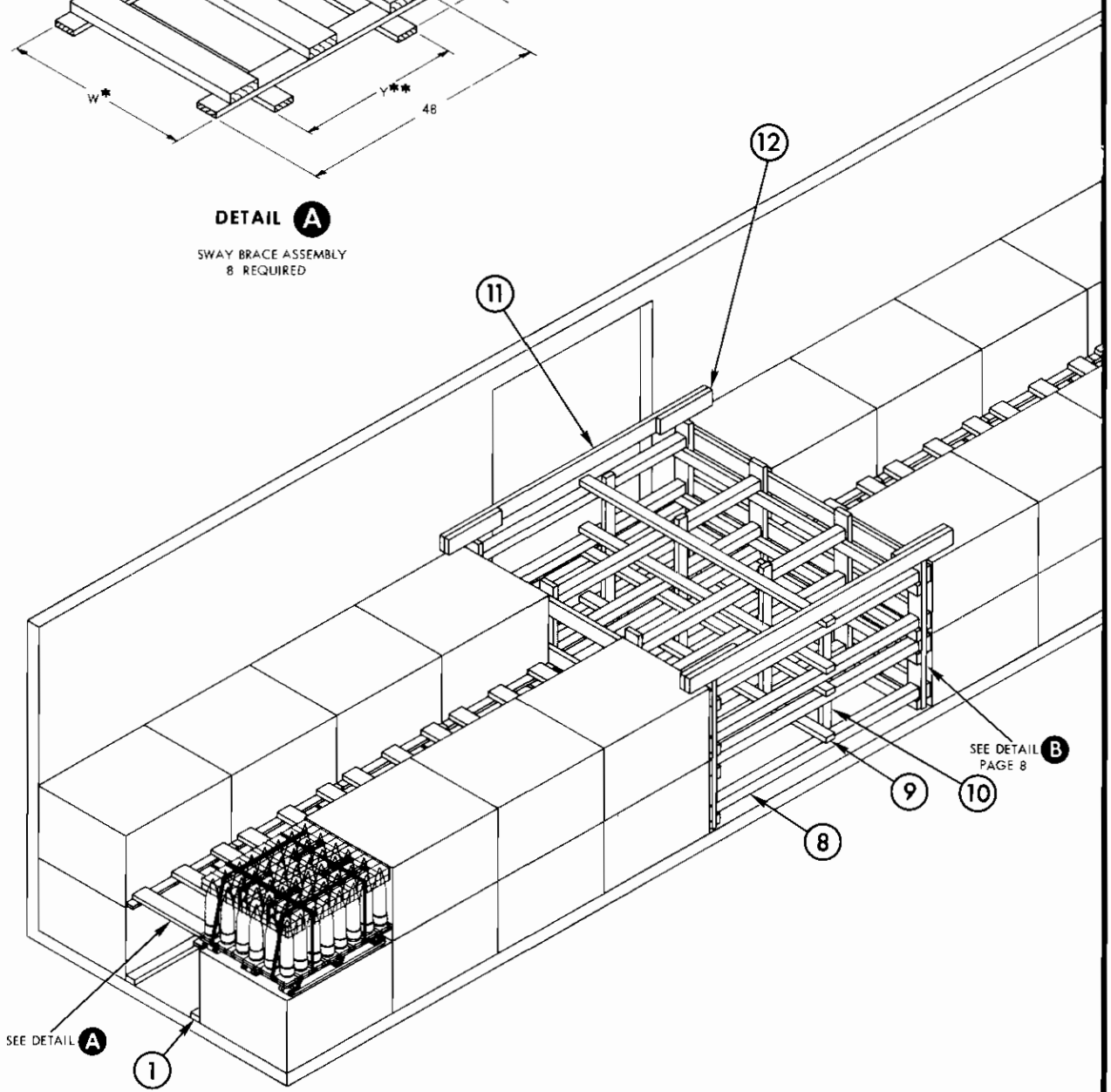


TABLE III

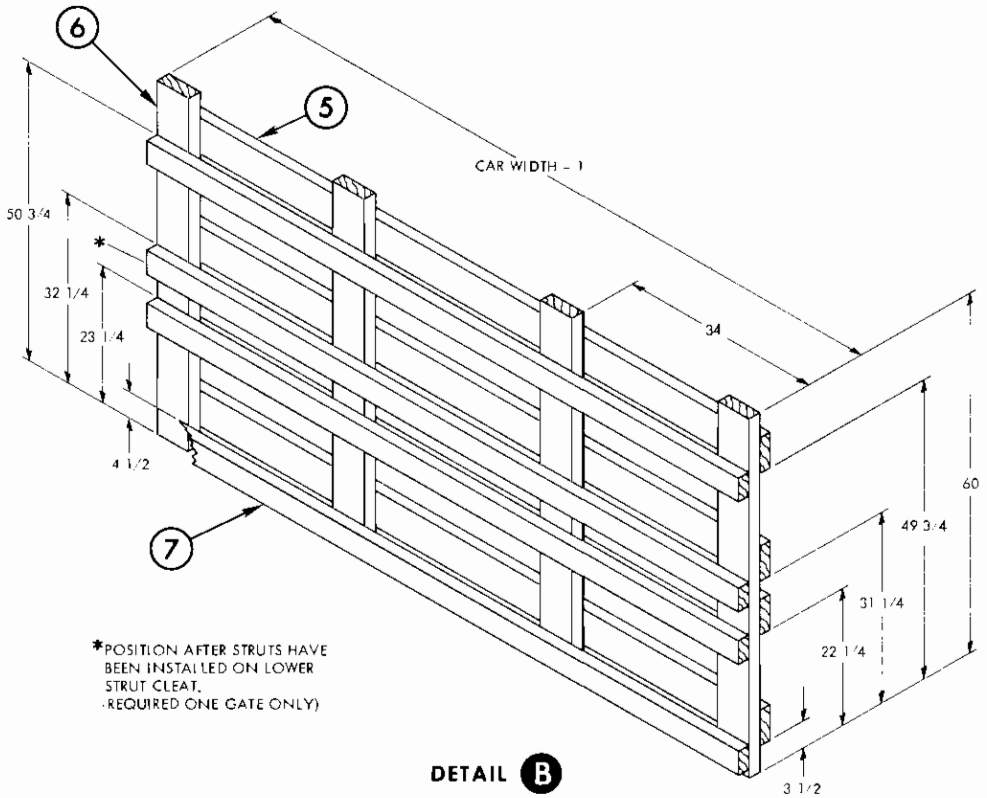
ITEM	CARLOAD WEIGHT (LBS) (INCLUDES 1263 LBS DUNNAGE WT) PALLET ADAPTER MK 11	
	MOD 0	MOD 1
PROJECTILE WITHOUT WATERPROOF PROTECTING CAP	96,143	94,383
PROJECTILE WITH WATERPROOF PROTECTING CAP	99,983	98,223
PROJECTILE WITH WINDSHIELD	96,143	94,383

CARLOAD DATA

NUMBER OF UNIT LOADS 32

CARLOAD WEIGHT . . . . . SEE TABLE III

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.



DETAIL B

CENTER GATE
2 REQUIRED

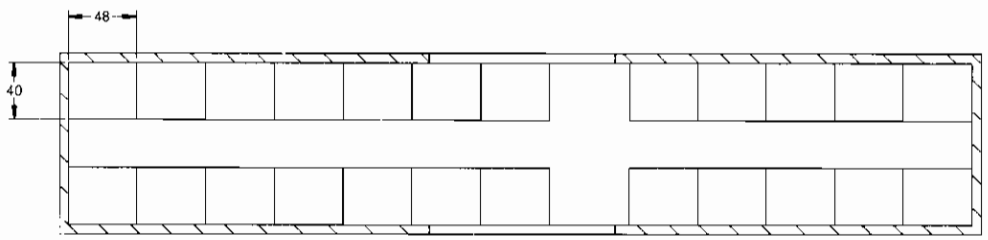
Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF EITHER 48 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING, OR 44 UNIT LOADS WHICH MAY BE LOADED AS SHOWN IN THE ALTERNATE LOADING PLAN ON PAGE 12. FOR THE ALTERNATE LOADING PLAN HORIZONTAL AND VERTICAL TIE BARS M. ST BE INSTALLED IN STRUTS IN DOORWAY AREA AS SHOWN ON PAGE 7.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 9 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 9 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A STRAPPING BOARD AND CROSS BRACE, TWO PER STACK (SEE DETAIL D) MUST BE INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. STRAP IS SECURED TO STRAPPING BOARD WITH TWO 1 1/4" STAPLES. DOORWAY PROTECTION, PIECES 11 AND 12 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



LOADING PLAN

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
12	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	11	2 PER JOINT	10d
11	VERTICAL DOORWAY MEMBER	2 x 3 x 72	4	DOOR POST	2 PER FOOT	20d
10	HOLDDOWN CLEAT	2 x 6 x 60	2	9	2 PER FOOT	10d
9	GATE HOLDDOWN	2 x 6 x 14 FT	2	CAR WALL	5 EACH END	10d
8	STRUT	4 x 4 x WEDGE FIT**	16	6	2 PER JOINT	16d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH -1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 60	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH -1	8	6	3 PER JOINT	10d
4	CROSS BRACE	2 x 6 x W*	36	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 48	24	2,4	SEE NOTE**	
2	SWAY BRACE SUPPORT	1 x 4 x (W + 8)*	24	SEE 3	-	-
1	SLEEPER	2 x 4 x 48	24	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS & NAILING DATA

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

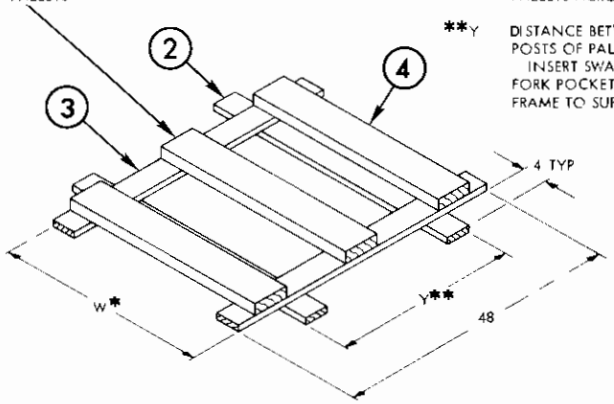
**2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4's.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

CROSS BRACES MUST BE CENTERED AGAINST LEGS OF PALLETS

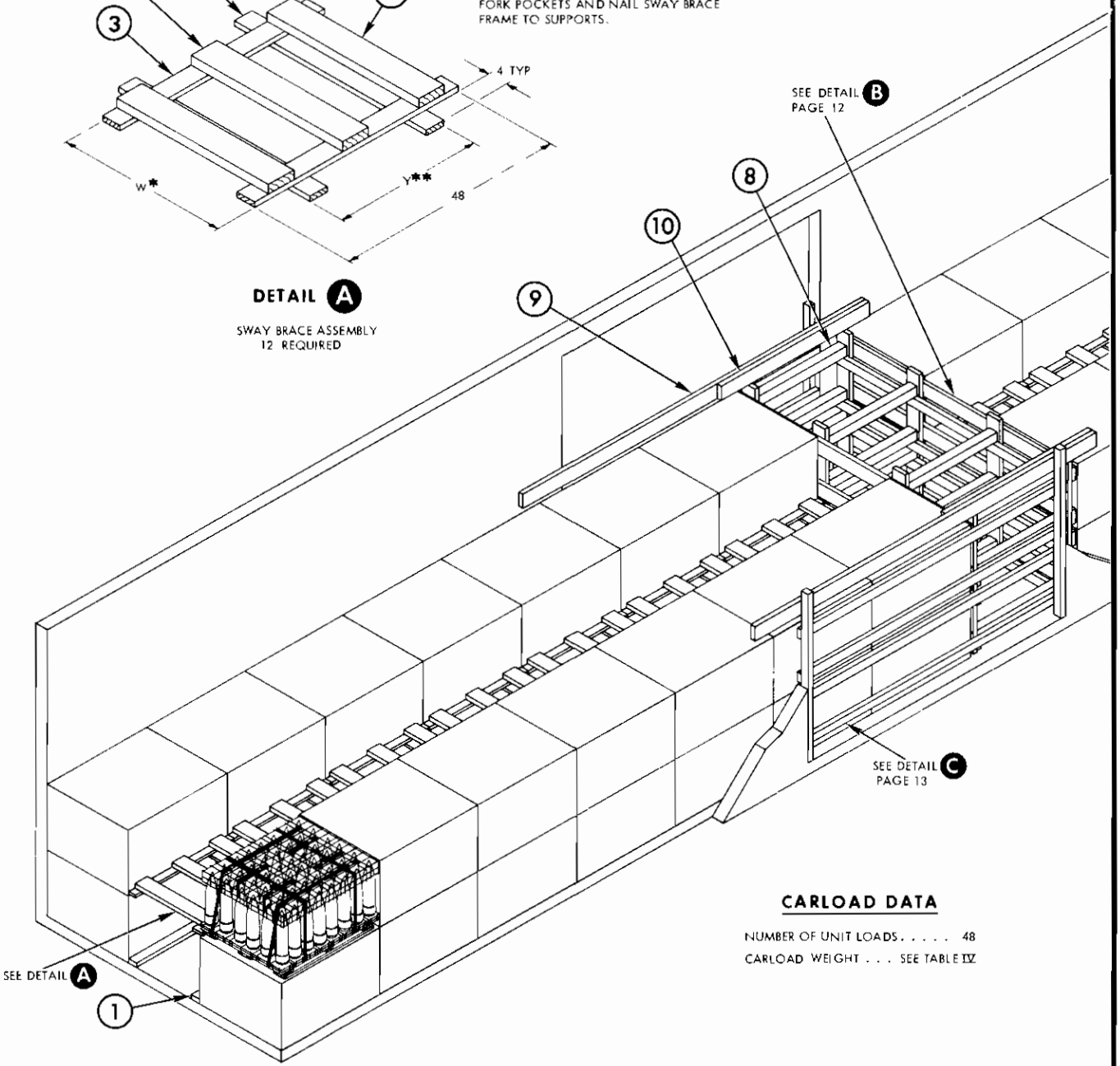
*W = DISTANCE BETWEEN POSTS OF PALLETS ACROSS THE CAR

**Y DISTANCE BETWEEN INSIDE OF OUTER POSTS OF PALLET. INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.



DETAIL A

SWAY BRACE ASSEMBLY
12 REQUIRED



CARLOAD DATA

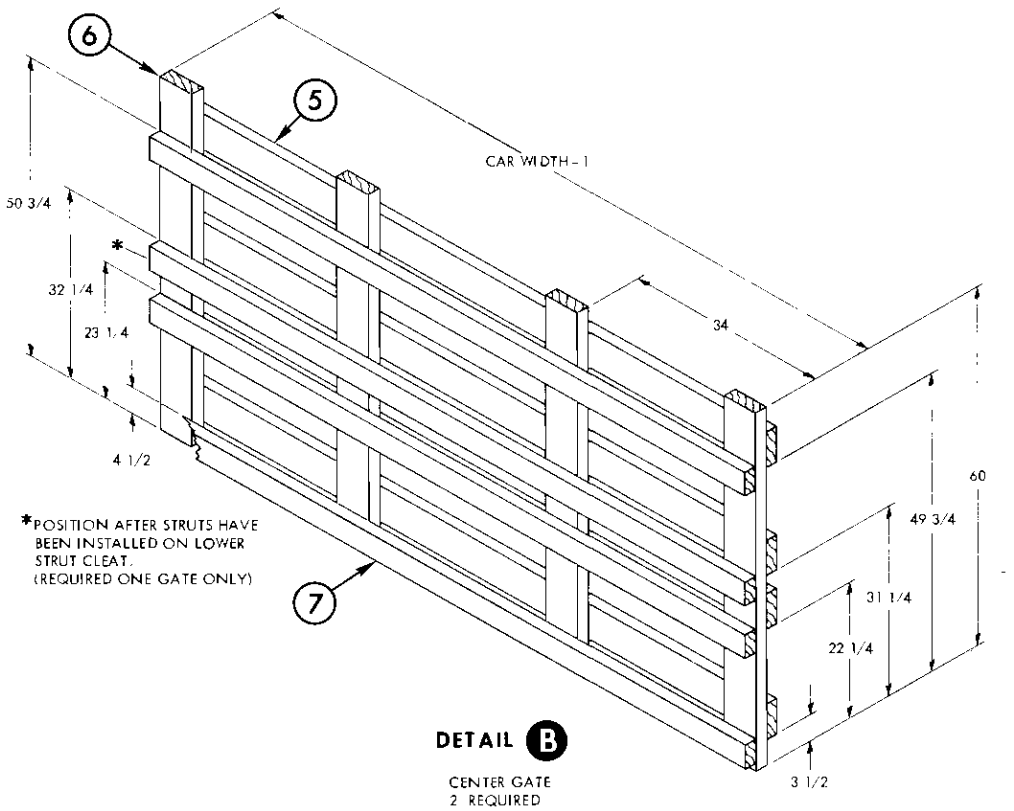
NUMBER OF UNIT LOADS 48

CARLOAD WEIGHT . . . SEE TABLE IV

TABLE IV

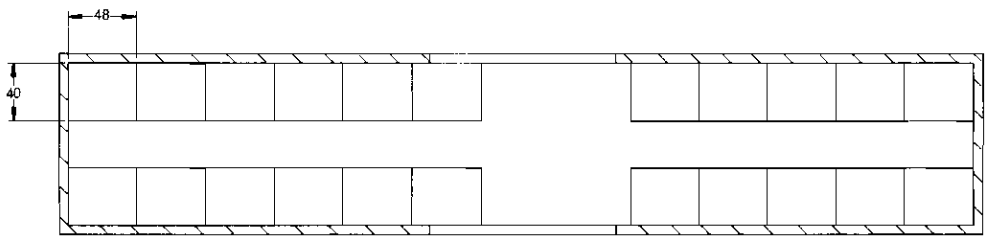
ITEM	CARLOAD WEIGHT (LBS) (INCLUDES 1284 LBS DUNNAGE WT) PALLET ADAPTER MK 11	
	MOD 0	MOD 1
PROJECTILE WITHOUT WATERPROOF PROTECTING CAP	143,604	140,964
PROJECTILE WITH WATERPROOF PROTECTING CAP	149,364	146,724
PROJECTILE WITH WINDSHIELD	143,604	140,964

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.



DETAIL B

CENTER GATE
2 REQUIRED



ALTERNATE LOADING PLAN

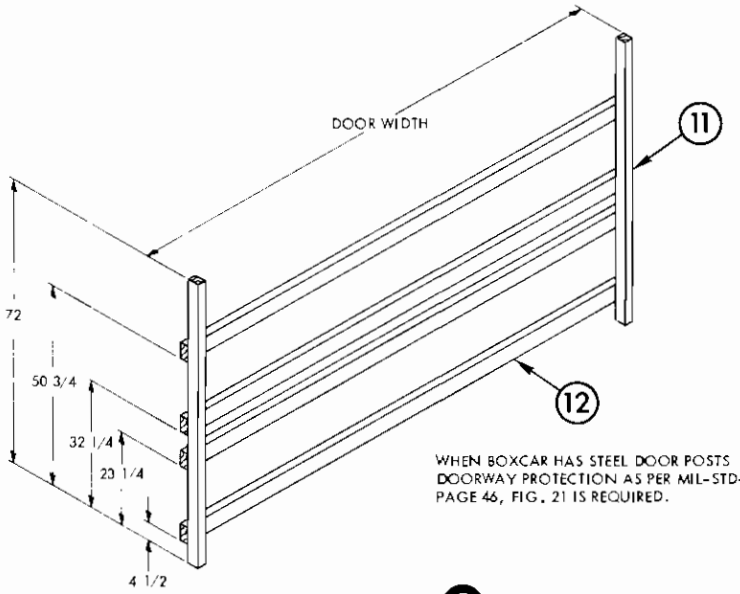
ALTERNATE CARLOAD DATA

NUMBER OF UNIT LOADS 44
CARLOAD WEIGHT . . . . . SEE TABLE V

TABLE V

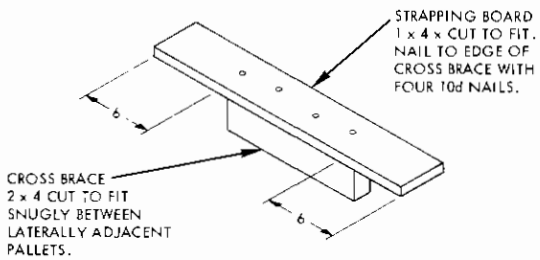
ITEM	CARLOAD WEIGHT (LBS) (INCLUDES 1400 LBS DUNNAGE WT) PALLET ADAPTER MK 11	
	MOD 0	MOD 1
PROJECTILE WITHOUT WATERPROOF PROTECTING CAP	131,860	129,440
PROJECTILE WITH WATERPROOF PROTECTING CAP	137,140	134,720
PROJECTILE WITH WINDSHIELD	131,860	129,440

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.



DETAIL C

DOORWAY PROTECTION
2 REQUIRED



DETAIL D

TOP SWAY BRACE

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.